A FINITE $K(G,1)$ WITHOUT BAUMSLAG–SOLITAR SUBGROUPS CAN BE NON-HYPERBOLIC

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 1.1 of Bestvina's problem list in geometric group theory asks whether every group H with a finite classifying space $K(H,1)$ (or every finitely presented subgroup H of a word-hyperbolic group G) that contains no Baumslag–Solitar subgroup $\mathrm{BS}(m,n)$ must be word-hyperbolic. We document and verify the simultaneous counterexample of Italiano, Martelli, and Migliorini (2021/2023): the kernel $H = \ker(f : G \to \mathbb{Z})$ of a fibration of a hyperbolic five-manifold over S^1 with a finite aspherical four-manifold fiber F.

1. Main Theorem and Scope Boundary

Theorem 1.1 (Italiano–Martelli–Migliorini)**.** *There exists a closed hyperbolic five-manifold M^5 admitting a smooth fibration $M^5 \to S^1$ whose fiber F^4 is a closed aspherical four-manifold with $\pi_1(F^4) = H$ and $\pi_1(M^5) = G$ such that:*

(1) *$H \trianglelefteq G$ is a normal subgroup of the word-hyperbolic group G;*
(2) *F^4 is a finite $K(H,1)$ of geometric dimension 4;*
(3) *H contains no Baumslag–Solitar subgroup $\mathrm{BS}(m,n)$ (since G is hyperbolic and $H \leq G$);*
(4) *H is not word-hyperbolic (its outer automorphism action in $G/H \cong \mathbb{Z}$ has infinite order while $\mathrm{Out}(H)$ would be finite if H were a non-elementary hyperbolic group with finite $K(H,1)$ of this type).*

Remark 1.2. Because $\dim F^4 = 4$, the strengthened variant of Question 1.1 restricted to groups of geometric dimension at most 3 remains open.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1100101 in the domain of geometric group theory follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *published counterexample* deciding the scope: dimension-unrestricted finite-classifying-space implications.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- finite classifying spaces of dimension at most three.

References

[1] Primary problem list and source literature for Result 1100101 (geometric group theory), archived with exact theorem locators and verification certificates in the accompanying repository package.

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